Akoizo Original Practice - 2027 Rotation Form 1

Division B | Remote Sensing | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Classify Sentinel-1 SAR as active or passive and explain the distinction.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
_____
_____

2. (2 points) Which kind of resolution describes the ground area represented by a pixel?

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Answer: _____
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3. (2 points) Does a shorter orbital period necessarily give an equally short revisit time for a specified ground point? Explain.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
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4. (2 points) Using c=3.0e8 m/s, find the frequency of light with wavelength 6.0e-7 m.

Answer: _____
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5. (2 points) Use Wien's law with b=2.90e-3 m K to find peak wavelength at 290 K.

Answer: _____
_____
_____

6. (2 points) A radar echo takes 2.0e-5 s for its round trip. With c=3.0e8 m/s, calculate range.

Answer: _____
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_____

7. (2 points) Find the changed patch east-west ground length.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
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8. (2 points) Find the changed patch north-south ground length.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
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9. (2 points) Find its ground area.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
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10. (2 points) Find its perimeter.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
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11. (2 points) Cloud obscures the optical image but the SAR image shows a surface pattern. Explain why these observations differ.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
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12. (2 points) A dark SAR patch is labeled water from brightness alone. Give one alternative explanation and a stronger check.

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Answer: _____
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13. (2 points) A classified map labels 41 of 100 equal-area pixels as forest. Calculate forest cover percentage.

Answer: _____
_____
_____

14. (2 points) Forest pixels decline from 81 to 51. How many equal-area pixels of forest are lost?

Answer: _____
_____
_____

15. (2 points) Explain why comparing a wet-season image with a dry-season image can mimic land-cover change.

Answer: _____
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16. (2 points) A topographic profile rises 60 m over 1,200 m horizontal distance. Find gradient as a percentage.

Answer: _____
_____
_____

17. (2 points) Two contours are 20 m apart vertically. Adjacent contours are closer horizontally in area A than area B. Which area is steeper and why?

Answer: _____
_____
_____

18. (2 points) A false-color display makes healthy vegetation red. Does that mean the vegetation reflects red light most strongly? Explain.

Answer: _____
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_____

19. (2 points) An ECOSTRESS thermal image shows a warmer paved area than a nearby canopy. Give an observation and a cautious interpretation.

Answer: _____
_____
_____

20. (2 points) A classifier was trained using labeled examples of water, forest and pavement. Is it supervised or unsupervised? Explain.

Answer: _____
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